

Algebra: Properties of Logarithms

Practice Worksheet • numberbender.com

N

Name: _____ Date: _____

Score: / 10

DIRECTIONS

Evaluate or simplify using Product, Quotient, and Power Rules of logarithms.

1 Evaluate:

$$\log_{10} 1000$$

Answer: _____

2 Apply log properties to simplify:

$$b^{\log_b 5}$$

Answer: _____

3 Apply log properties to simplify:

$$\log_b b^7$$

Answer: _____

4 Apply log properties to simplify:

$$\log_b 1$$

Answer: _____

5 Evaluate:

$$\log_2 1$$

Answer: _____

6 Evaluate:

$$\log_5 125$$

Answer: _____

7 Evaluate:

$$\log_7 7$$

Answer: _____

8 Apply log properties to simplify:

$$\log_{10} 10^x$$

Answer: _____

9 Evaluate:

$$\log_2 32$$

Answer: _____

10 Evaluate:

$$\ln e^5$$

Answer: _____



Properties of Logarithms Worksheet

More free worksheets, video lessons & courses:

numberbender.com/math-worksheets

Math is hard. So we made it human.
Labanan ang agwat.

Free to print & share (CC BY 4.0)

Page 1 of 2 · Video: youtu.be/Ljf38D-sGUU



Watch the lesson



More free sheets

Answer Key & Solutions

Algebra: Properties of Logarithms • Numberbender

N

TEACHER NOTES Key values: $\log_b(1)=0$, $\log_b(b)=1$, $\log_b(b^n)=n$. These appear frequently on exams.

1 Evaluate:

$$= 3$$

$$\log_{10} 1000$$

2 Apply log properties to simplify:

$$= 5$$

$$b^{\log_b 5}$$

3 Apply log properties to simplify:

$$= 7$$

$$\log_b b^7$$

4 Apply log properties to simplify:

$$= 0$$

$$\log_b 1$$

5 Evaluate:

$$= 0$$

$$\log_2 1$$

6 Evaluate:

$$= 3$$

$$\log_5 125$$

7 Evaluate:

$$= 1$$

$$\log_7 7$$

8 Apply log properties to simplify:

$$= x$$

$$\log_{10} 10^x$$

9 Evaluate:

$$= 5$$

$$\log_2 32$$

10 Evaluate:

$$= 5$$

$$\ln e^5$$

